

# Chapter 12

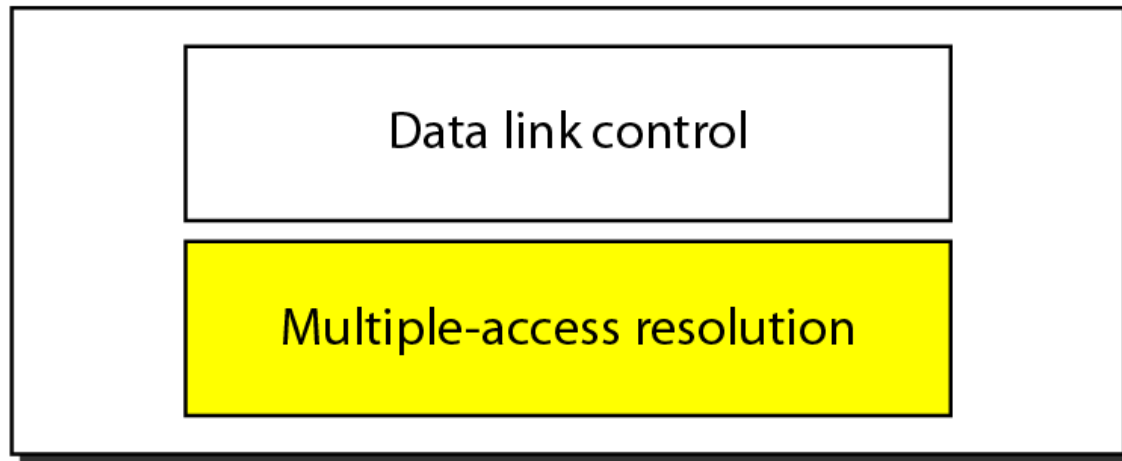
## Multiple Access

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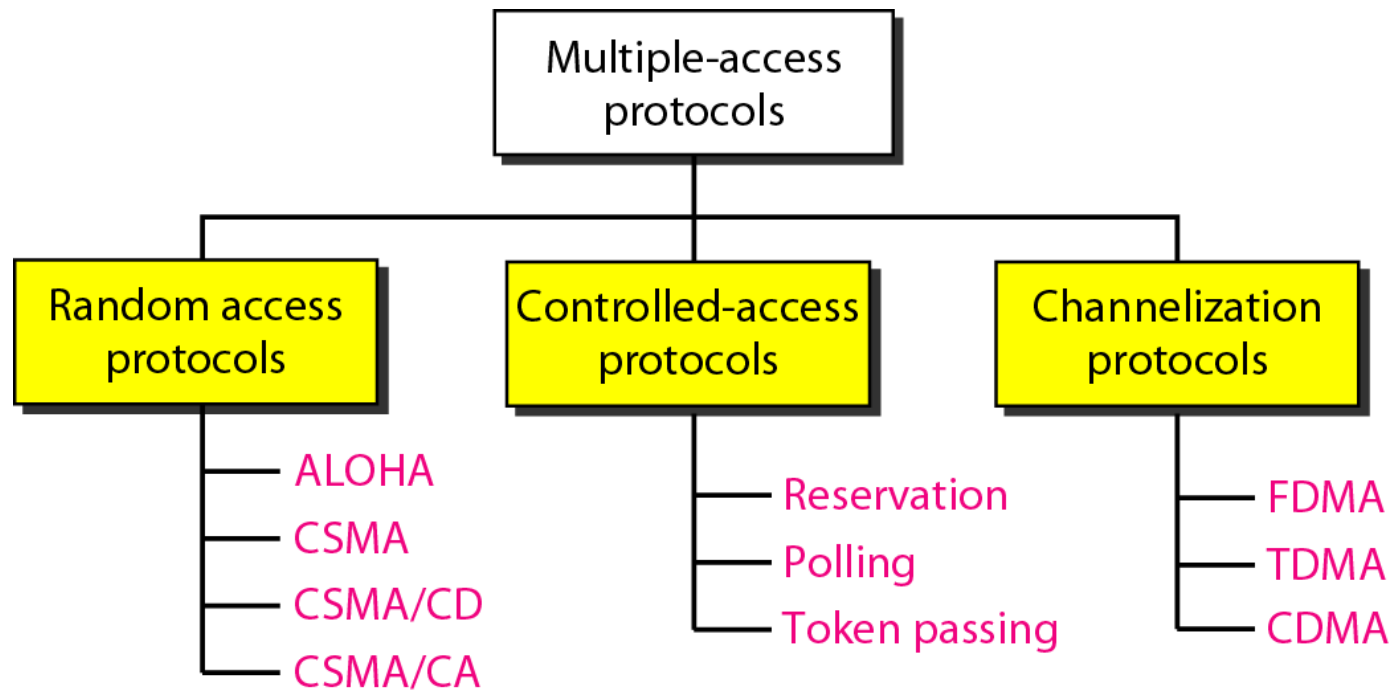
**Figure 12.1** *Data link layer divided into two functionality-oriented sublayers*

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Data link layer



**Figure 12.2** *Taxonomy of multiple-access protocols discussed in this chapter*



## 12-1 RANDOM ACCESS

*In **random access** or **contention** methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.*

*Topics discussed in this section:*

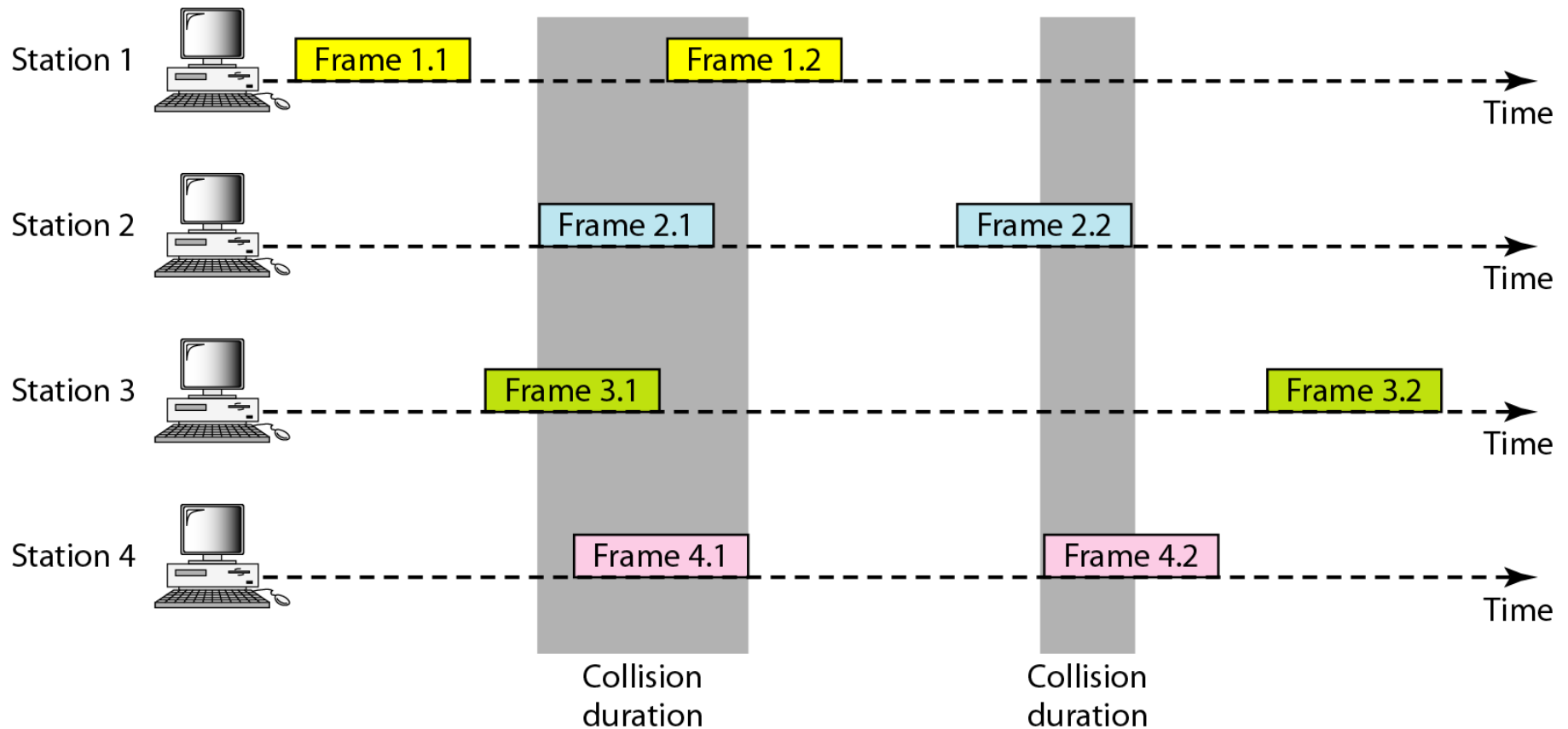
**ALOHA**

**Carrier Sense Multiple Access**

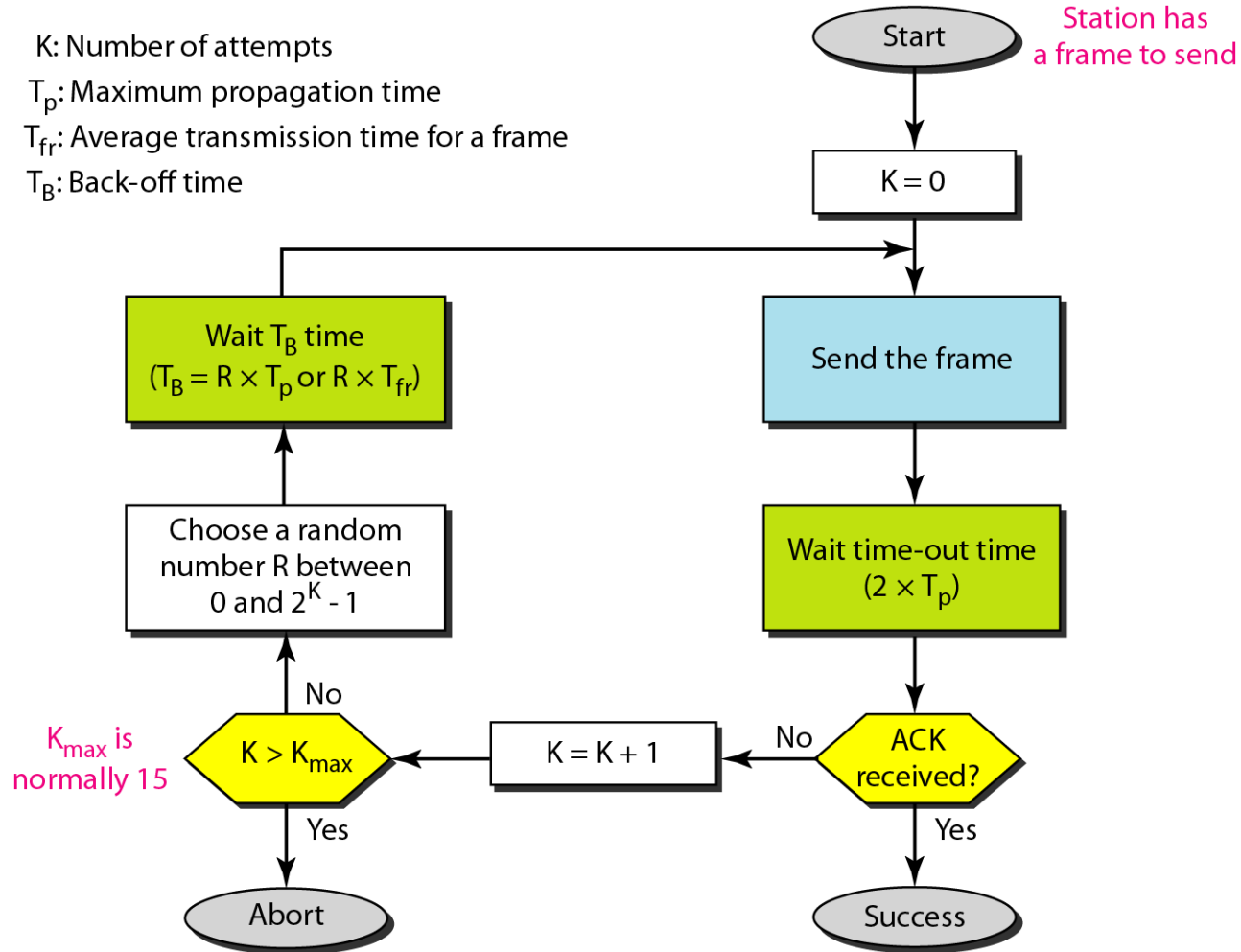
**Carrier Sense Multiple Access with Collision Detection**

**Carrier Sense Multiple Access with Collision Avoidance**

**Figure 12.3** *Frames in a pure ALOHA network*



**Figure 12.4** *Procedure for pure ALOHA protocol*





## Example 12.1

*The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at  $3 \times 10^8$  m/s, we find*

$$T_p = (600 \times 10^5) / (3 \times 10^8) = 2 \text{ ms.}$$

*Now we can find the value of  $T_B$  for different values of  $K$ .*

- a. For  $K = 1$ , the range is  $\{0, 1\}$ . The station needs to generate a random number with a value of 0 or 1. This means that  $T_B$  is either 0 ms ( $0 \times 2$ ) or 2 ms ( $1 \times 2$ ), based on the outcome of the random variable.*

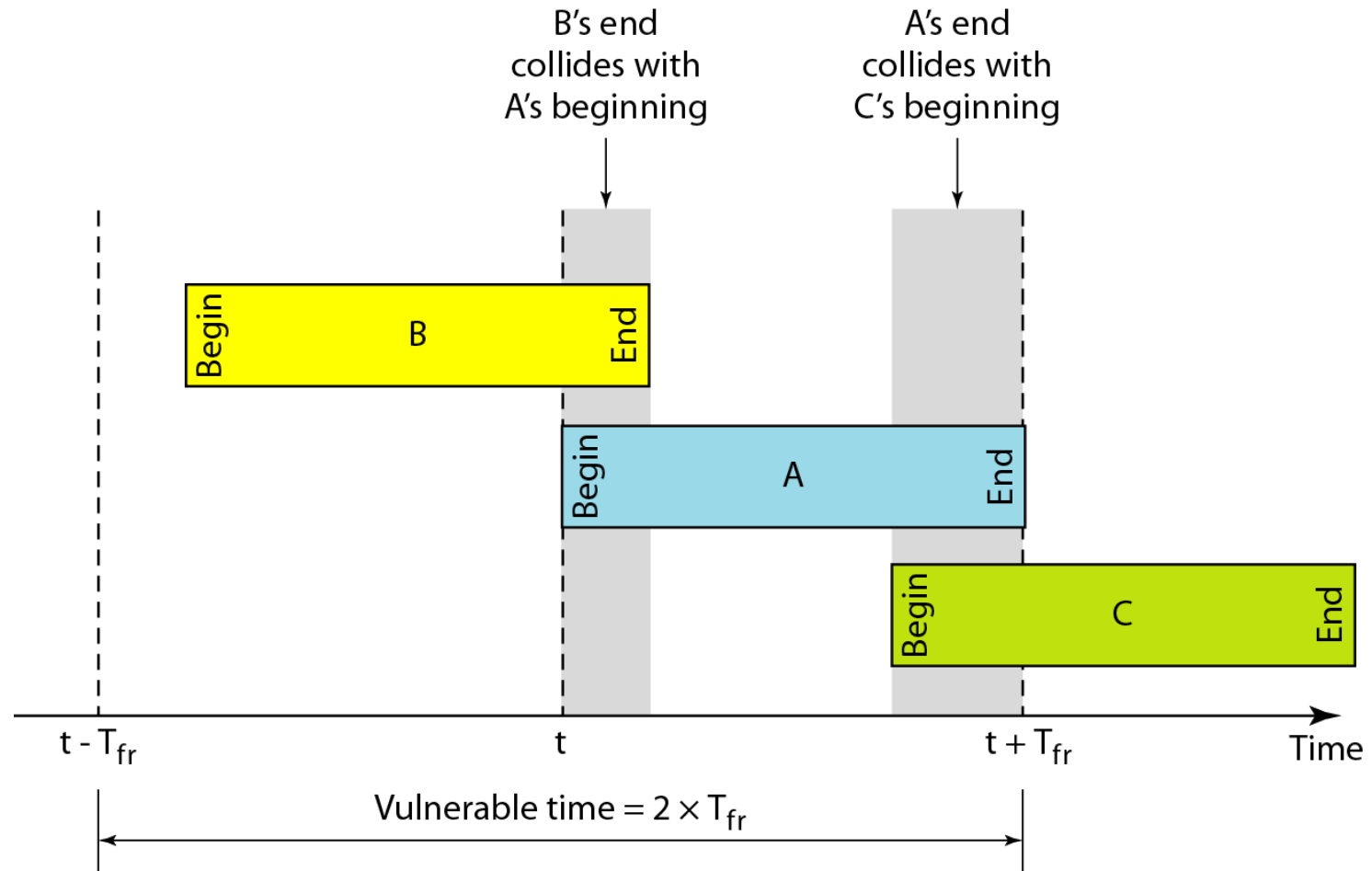


## *Example 12.1 (continued)*

- b. For  $K = 2$ , the range is  $\{0, 1, 2, 3\}$ . This means that  $T_B$  can be 0, 2, 4, or 6 ms, based on the outcome of the random variable.*
- c. For  $K = 3$ , the range is  $\{0, 1, 2, 3, 4, 5, 6, 7\}$ . This means that  $T_B$  can be 0, 2, 4, . . . , 14 ms, based on the outcome of the random variable.*
- d. We need to mention that if  $K > 10$ , it is normally set to 10.*



**Figure 12.5** *Vulnerable time for pure ALOHA protocol*





## Example 12.2

*A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?*

### **Solution**

*Average frame transmission time  $T_{fr}$  is 200 bits/200 kbps or 1 ms. The vulnerable time is  $2 \times 1 \text{ ms} = 2 \text{ ms}$ . This means no station should send later than 1 ms before this station starts transmission and no station should start sending during the one 1-ms period that this station is sending.*



*Note*

The throughput for pure ALOHA is

$$S = G \times e^{-2G} .$$

The maximum throughput

$$S_{\max} = 0.184 \text{ when } G = (1/2).$$

## Example 12.3

*A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces*

- a. 1000 frames per second    b. 500 frames per second*
- c. 250 frames per second.*

### *Solution*

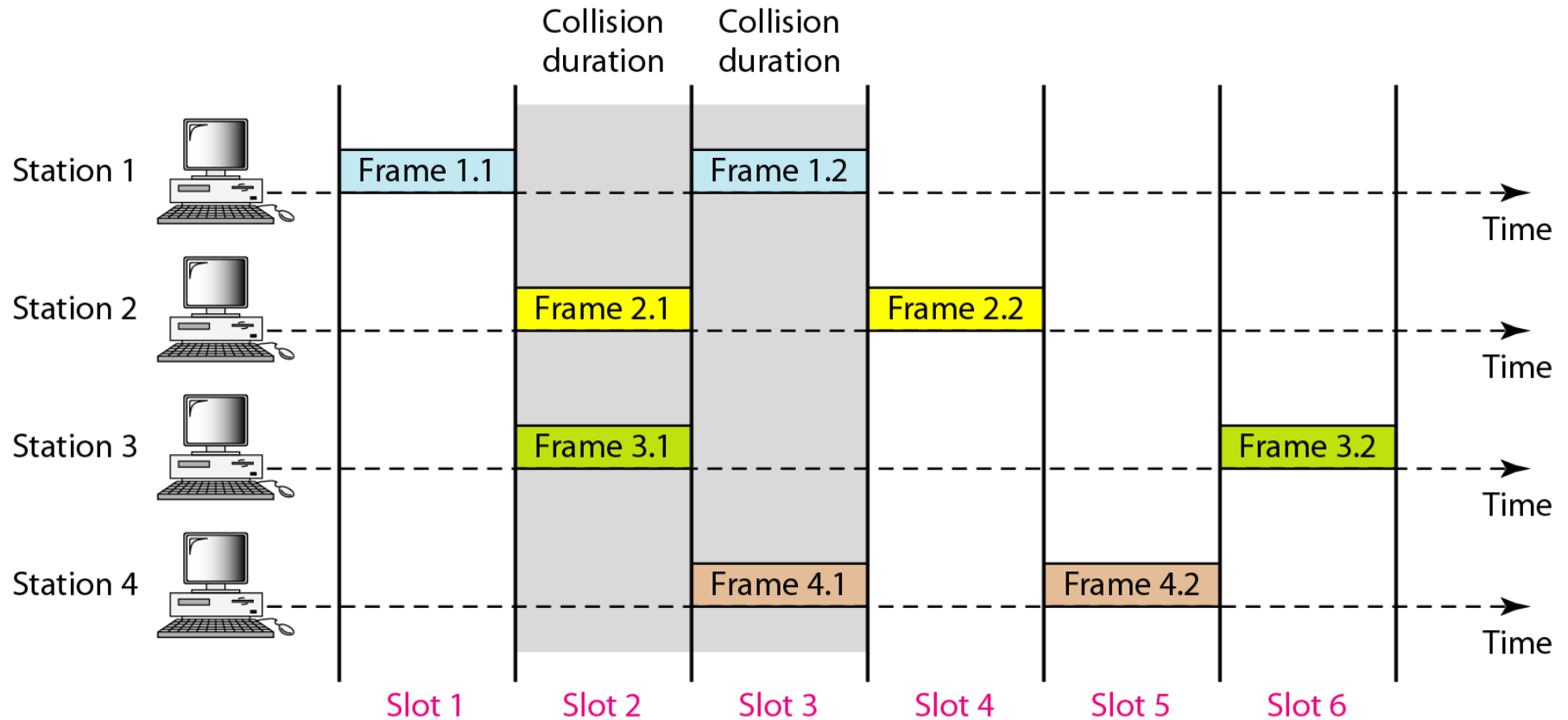
*The frame transmission time is 200/200 kbps or 1 ms.*

- a. If the system creates 1000 frames per second, this is 1 frame per millisecond. The load is 1. In this case  $S = G \times e^{-2G}$  or  $S = 0.135$  (13.5 percent). This means that the throughput is  $1000 \times 0.135 = 135$  frames. Only 135 frames out of 1000 will probably survive.*

## **Example 12.3 (continued)**

- b.** *If the system creates 500 frames per second, this is (1/2) frame per millisecond. The load is (1/2). In this case  $S = G \times e^{-2G}$  or  $S = 0.184$  (18.4 percent). This means that the throughput is  $500 \times 0.184 = 92$  and that only 92 frames out of 500 will probably survive. Note that this is the maximum throughput case, percentage-wise.*
- c.** *If the system creates 250 frames per second, this is (1/4) frame per millisecond. The load is (1/4). In this case  $S = G \times e^{-2G}$  or  $S = 0.152$  (15.2 percent). This means that the throughput is  $250 \times 0.152 = 38$ . Only 38 frames out of 250 will probably survive.*

**Figure 12.6** *Frames in a slotted ALOHA network*





*Note*

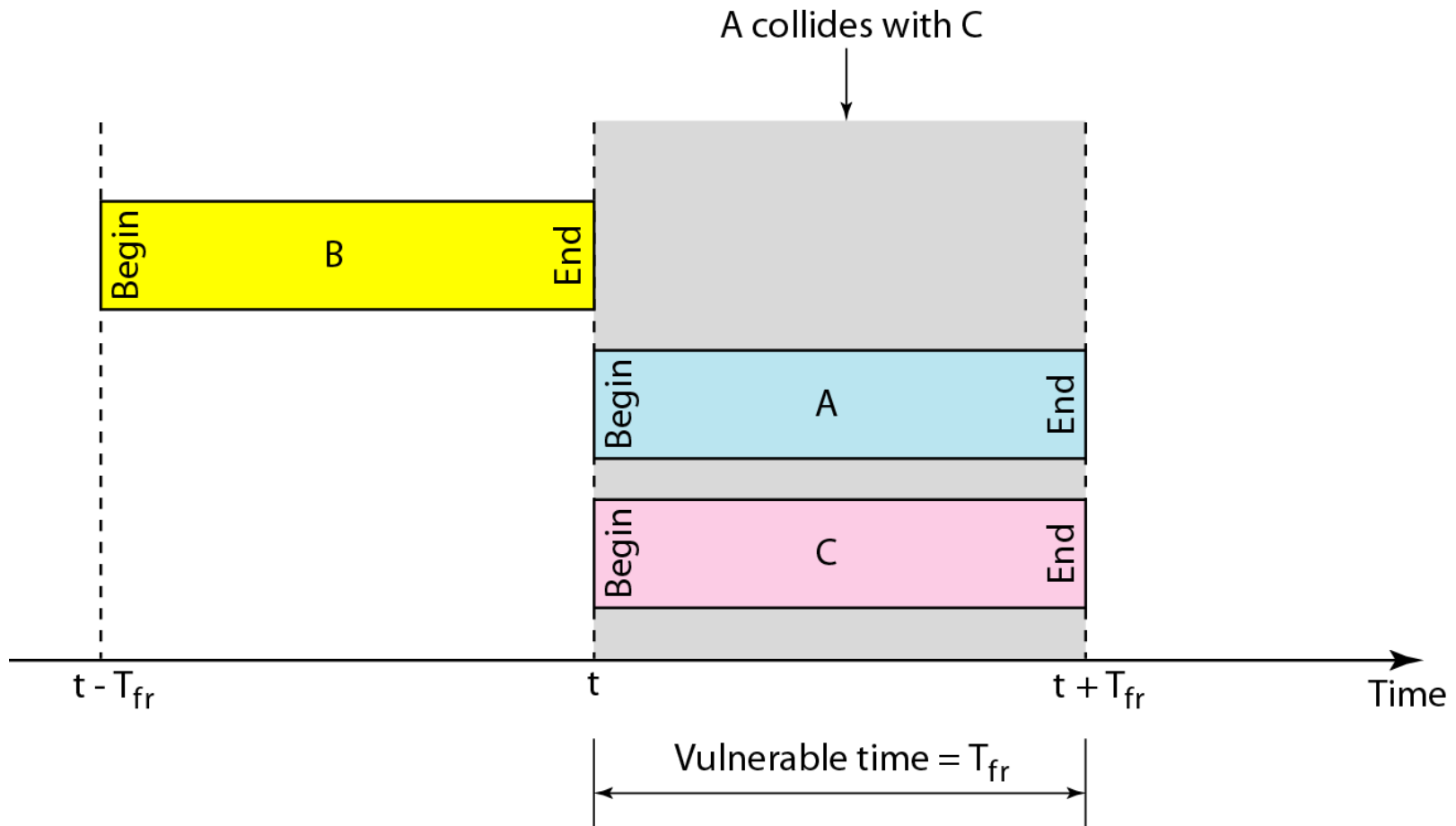
**The throughput for slotted ALOHA is**

$$S = G \times e^{-G} .$$

**The maximum throughput**

$$S_{\max} = 0.368 \text{ when } G = 1.$$

**Figure 12.7** *Vulnerable time for slotted ALOHA protocol*





## Example 12.4

*A slotted ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces*

- a. 1000 frames per second    b. 500 frames per second*
- c. 250 frames per second.*

### *Solution*

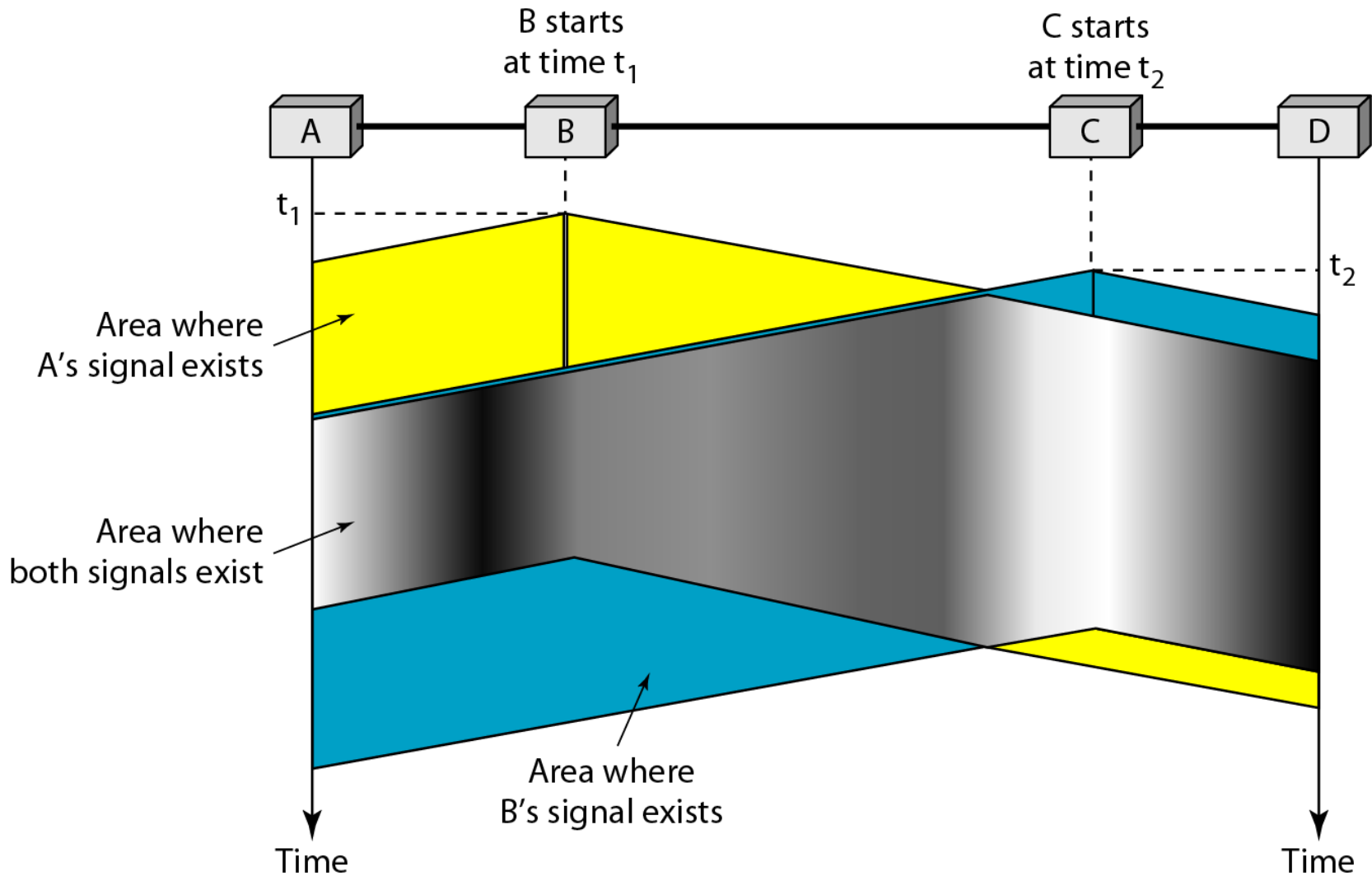
*The frame transmission time is 200/200 kbps or 1 ms.*

- a. If the system creates 1000 frames per second, this is 1 frame per millisecond. The load is 1. In this case  $S = G \times e^{-G}$  or  $S = 0.368$  (36.8 percent). This means that the throughput is  $1000 \times 0.0368 = 368$  frames. Only 386 frames out of 1000 will probably survive.*

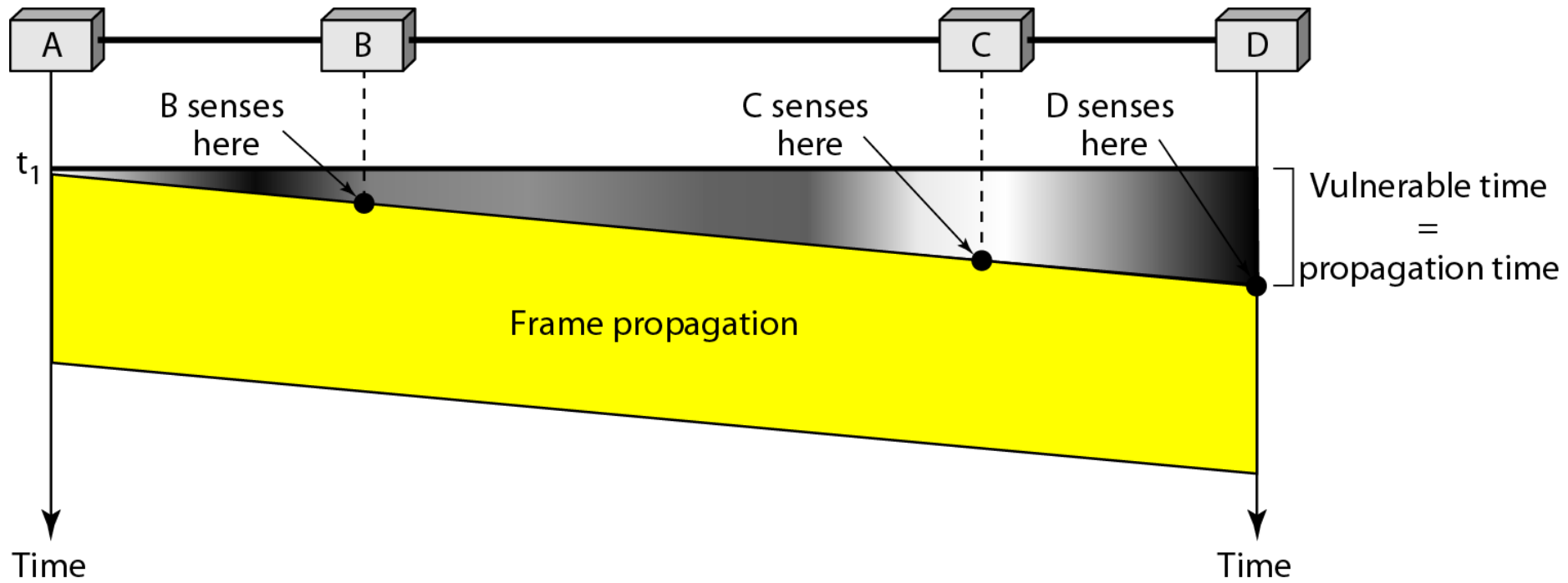
## **Example 12.4 (continued)**

- b.** *If the system creates 500 frames per second, this is (1/2) frame per millisecond. The load is (1/2). In this case  $S = G \times e^{-G}$  or  $S = 0.303$  (30.3 percent). This means that the throughput is  $500 \times 0.303 = 151$ . Only 151 frames out of 500 will probably survive.*
- c.** *If the system creates 250 frames per second, this is (1/4) frame per millisecond. The load is (1/4). In this case  $S = G \times e^{-G}$  or  $S = 0.195$  (19.5 percent). This means that the throughput is  $250 \times 0.195 = 49$ . Only 49 frames out of 250 will probably survive.*

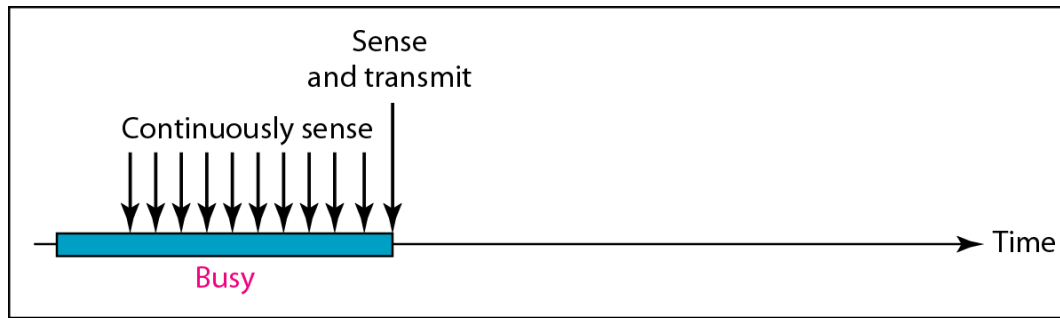
**Figure 12.8** *Space/time model of the collision in CSMA*



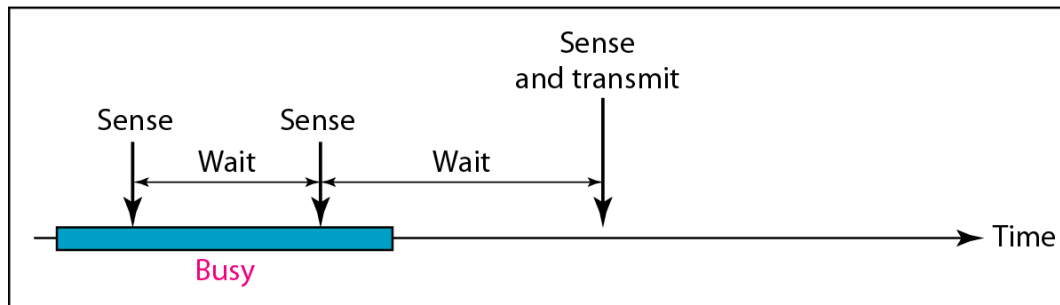
**Figure 12.9** *Vulnerable time in CSMA*



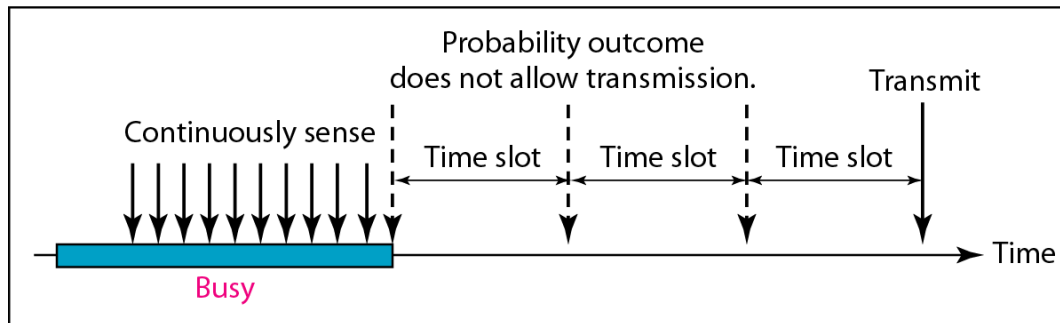
**Figure 12.10** *Behavior of three persistence methods*



a. 1-persistent

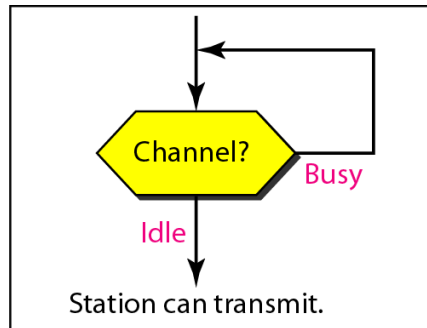


b. Nonpersistent

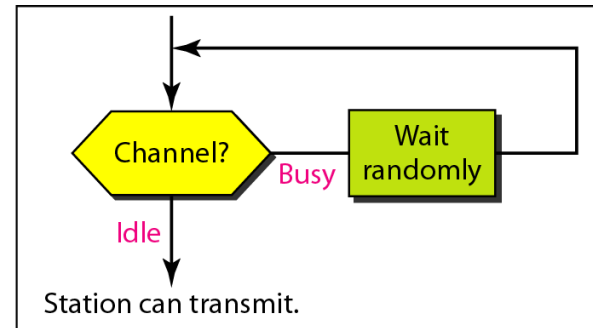


c. p-persistent

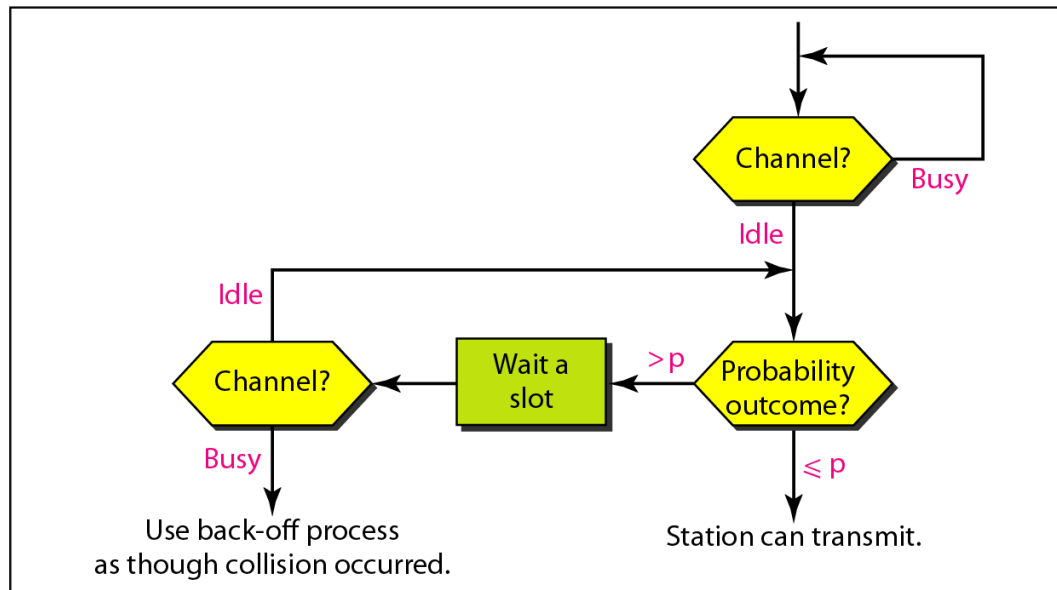
**Figure 12.11** *Flow diagram for three persistence methods*



a. 1-persistent

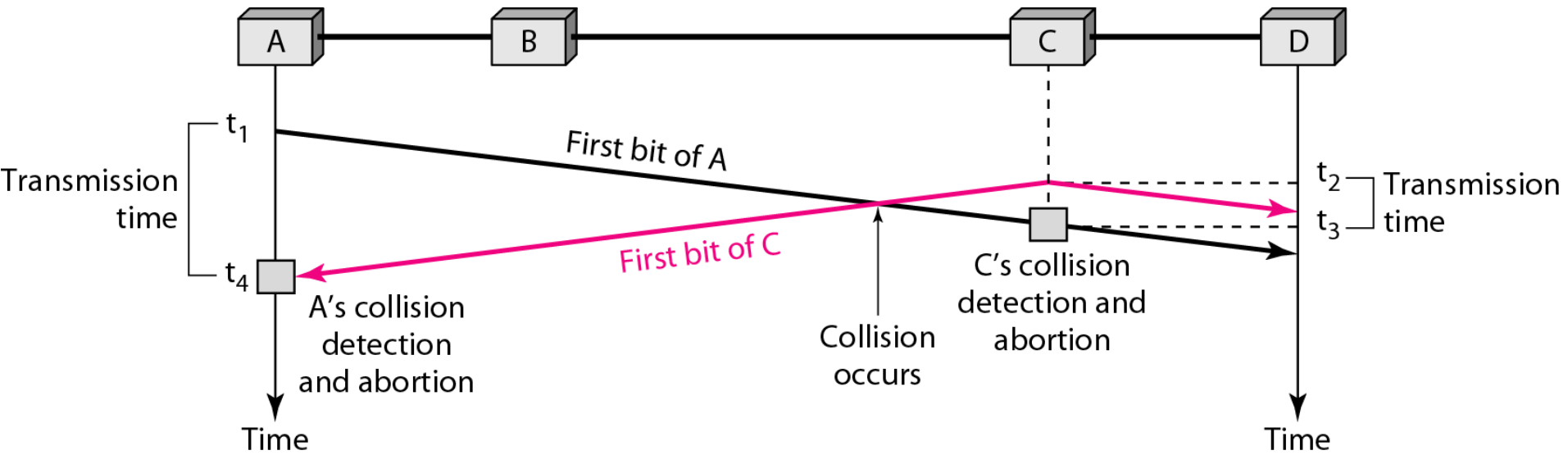


b. Nonpersistent

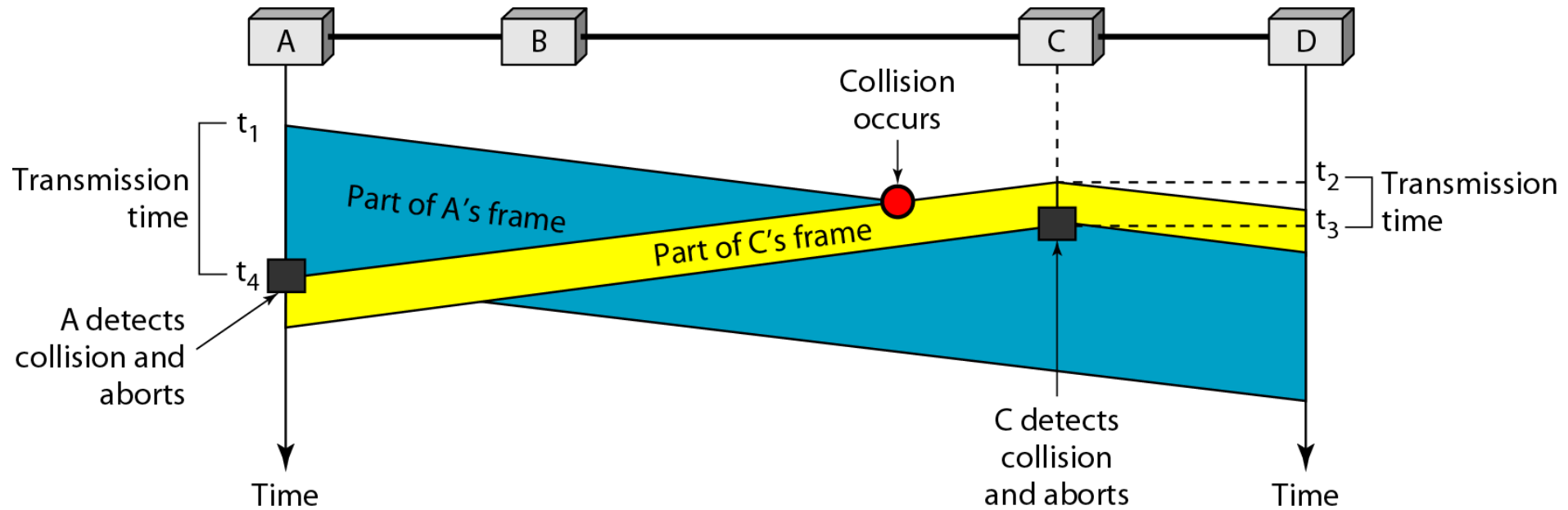


c. p-persistent

**Figure 12.12** *Collision of the first bit in CSMA/CD*



**Figure 12.13** *Collision and abortion in CSMA/CD*







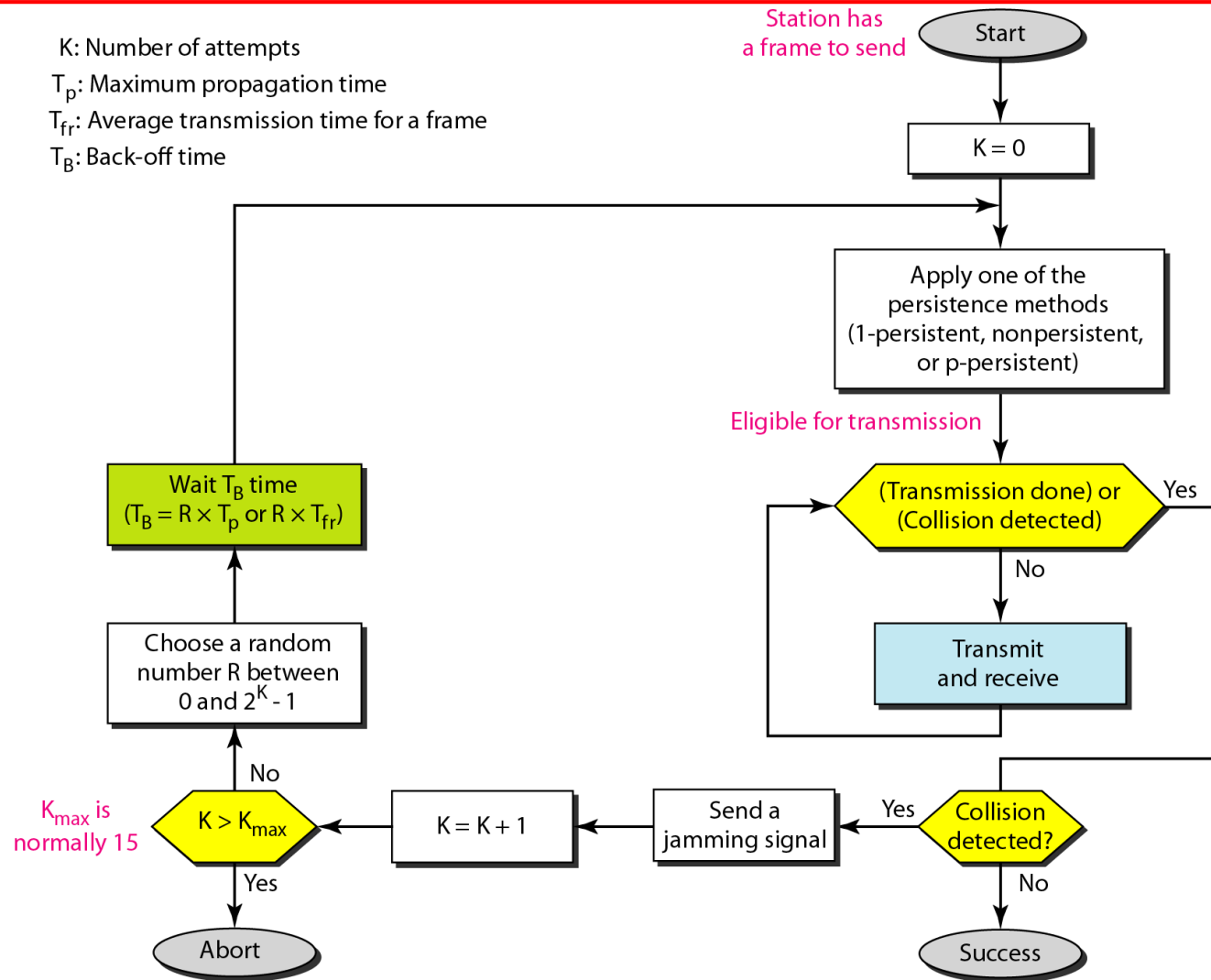
## Example 12.5

*A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including the delays in the devices and ignoring the time needed to send a jamming signal, as we see later) is  $25.6 \mu\text{s}$ , what is the minimum size of the frame?*

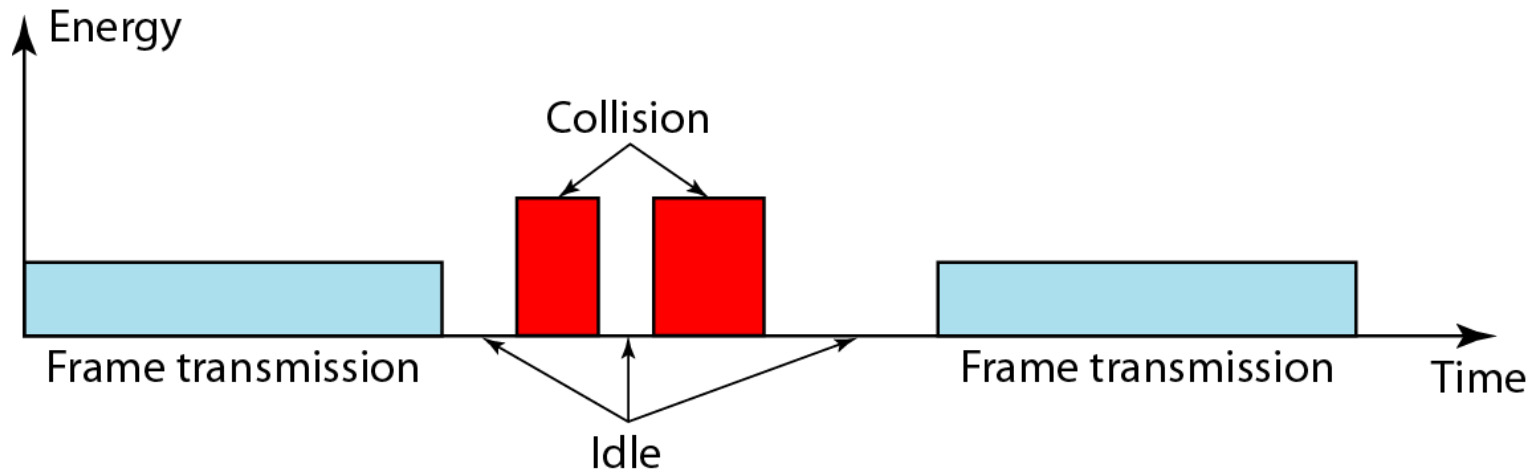
### **Solution**

*The frame transmission time is  $T_{fr} = 2 \times T_p = 51.2 \mu\text{s}$ . This means, in the worst case, a station needs to transmit for a period of  $51.2 \mu\text{s}$  to detect the collision. The minimum size of the frame is  $10 \text{ Mbps} \times 51.2 \mu\text{s} = 512$  bits or 64 bytes. This is actually the minimum size of the frame for Standard Ethernet.*

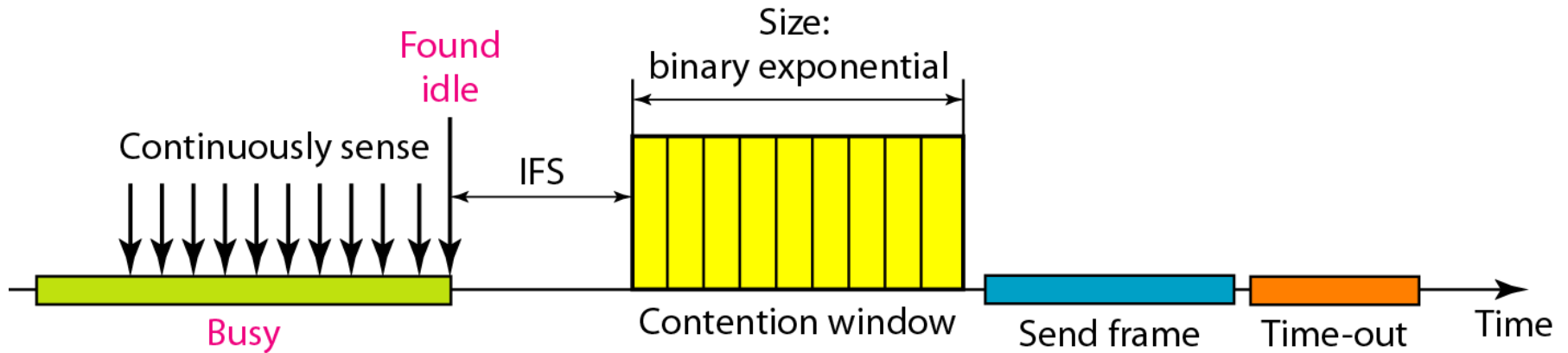
**Figure 12.14** *Flow diagram for the CSMA/CD*



**Figure 12.15** *Energy level during transmission, idleness, or collision*



**Figure 12.16** *Timing in CSMA/CA*





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*Note*

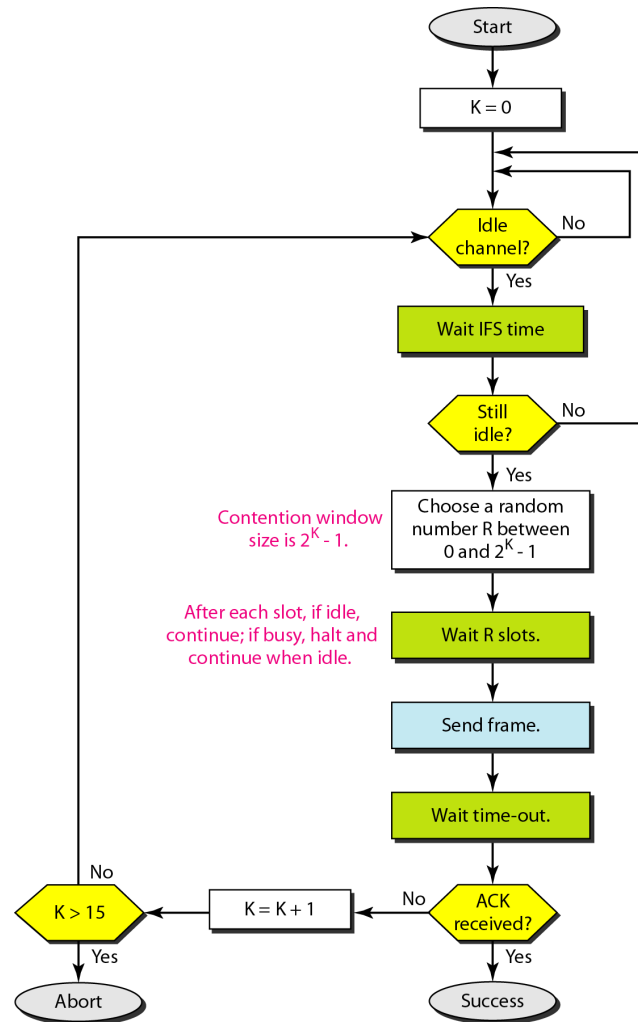
**In CSMA/CA, the IFS can also be used to define the priority of a station or a frame.**



*Note*

**In CSMA/CA, if the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.**

**Figure 12.17** *Flow diagram for CSMA/CA*



## 12-2 CONTROLLED ACCESS

*In **controlled access**, the stations consult one another to find which station has the right to send. A station cannot send unless it has been authorized by other stations. We discuss three popular controlled-access methods.*

*Topics discussed in this section:*

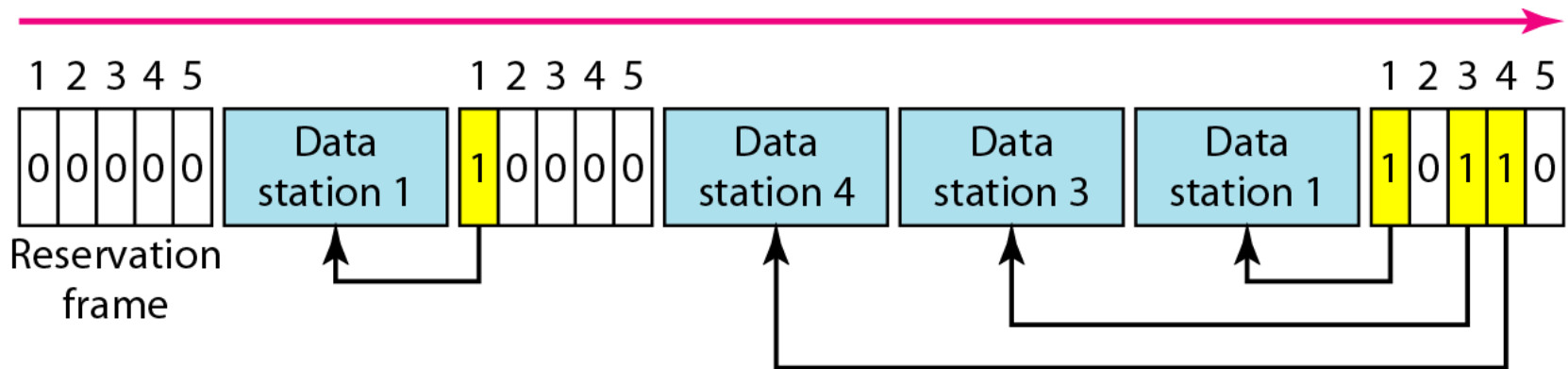
Reservation

Polling

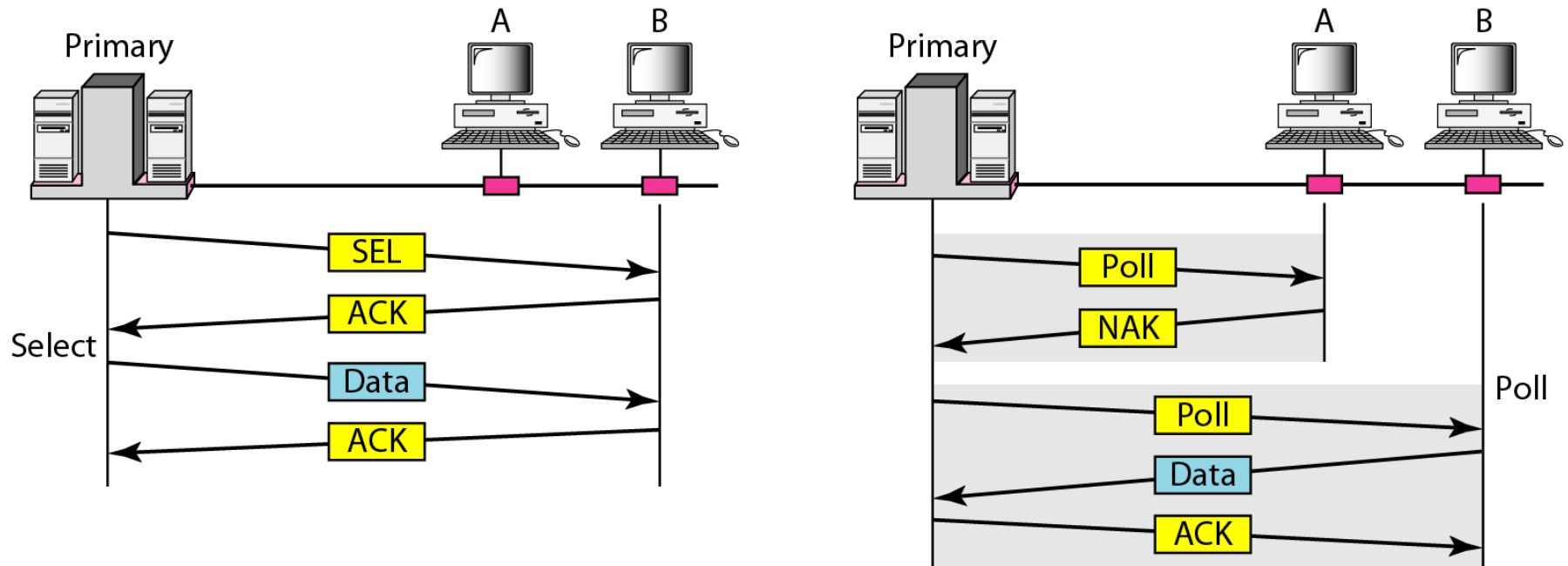
Token Passing



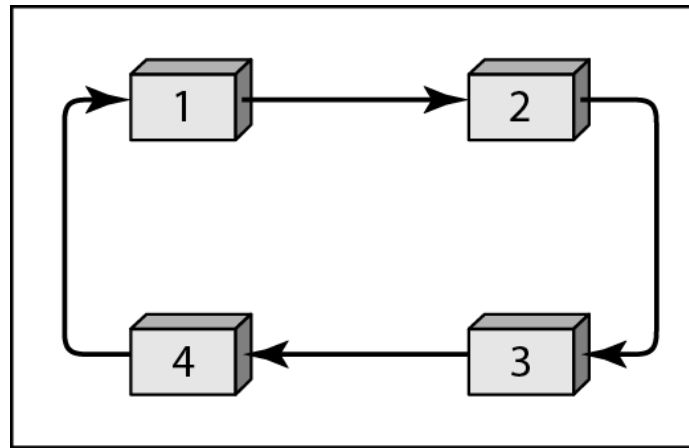
**Figure 12.18** *Reservation access method*



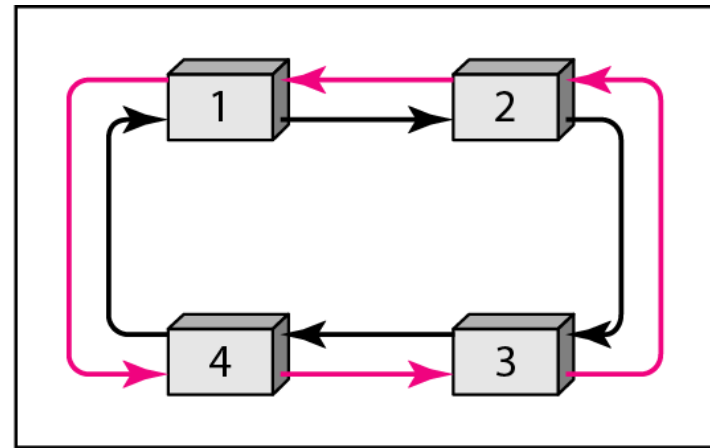
**Figure 12.19** *Select and poll functions in polling access method*



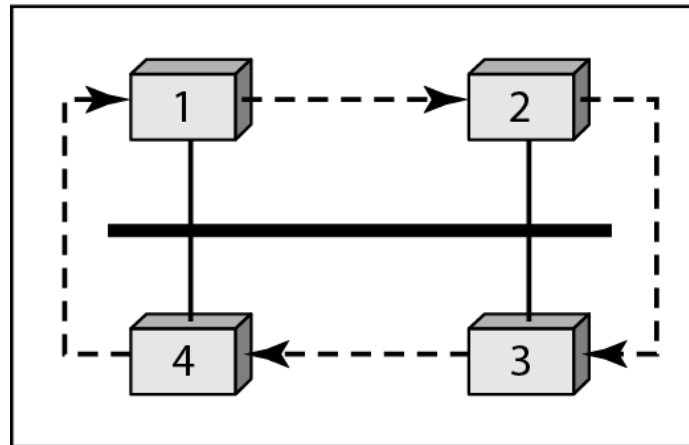
**Figure 12.20** *Logical ring and physical topology in token-passing access method*



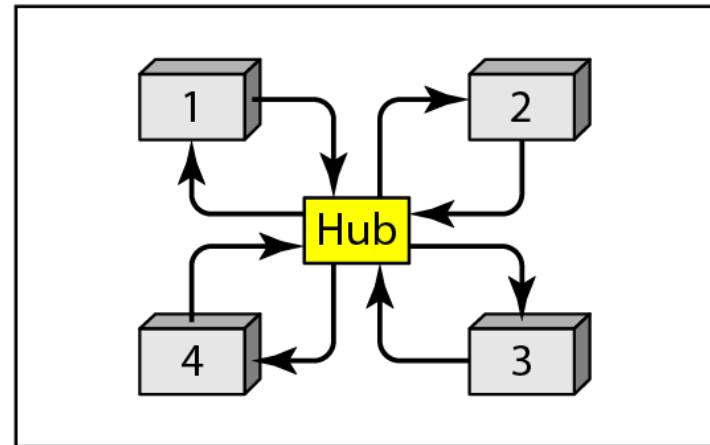
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

## 12-3 CHANNELIZATION

***Channelization*** is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations. In this section, we discuss three channelization protocols.

### **Topics discussed in this section:**

**Frequency-Division Multiple Access (FDMA)**

**Time-Division Multiple Access (TDMA)**

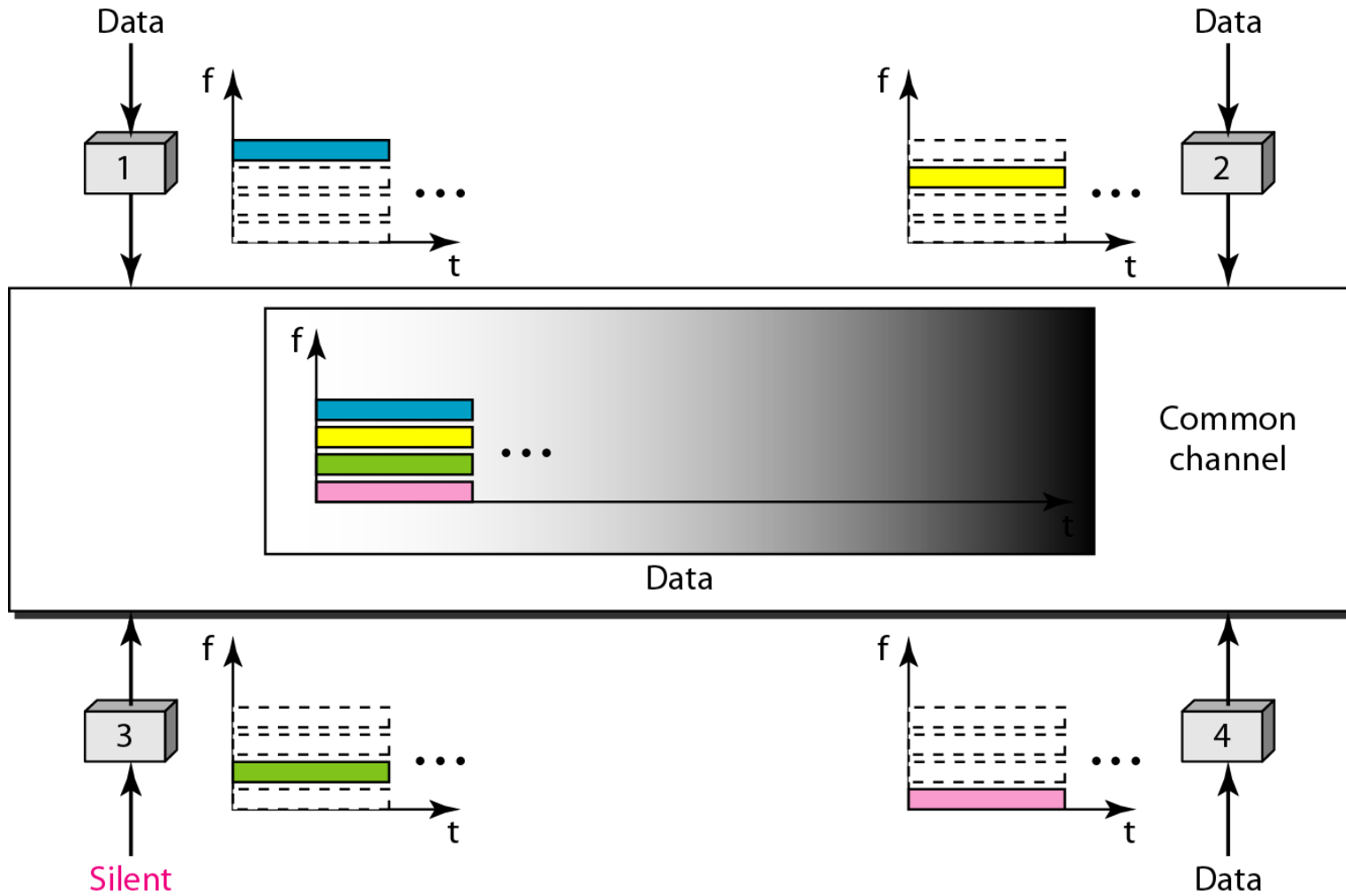
**Code-Division Multiple Access (CDMA)**



*Note*

**We see the application of all these methods in Chapter 16 when we discuss cellular phone systems.**

**Figure 12.21** *Frequency-division multiple access (FDMA)*

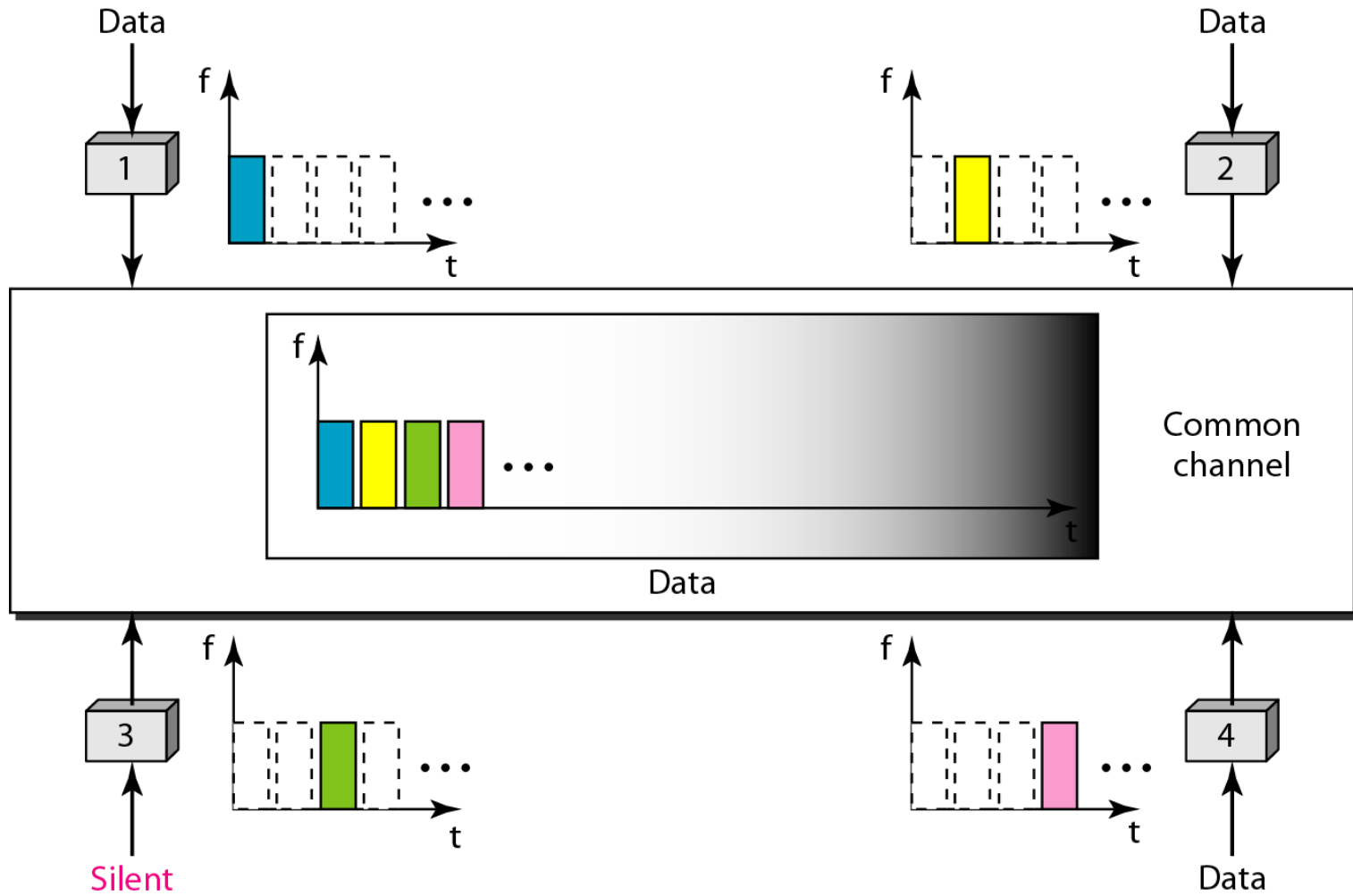




*Note*

**In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.**

**Figure 12.22** *Time-division multiple access (TDMA)*







*Note*

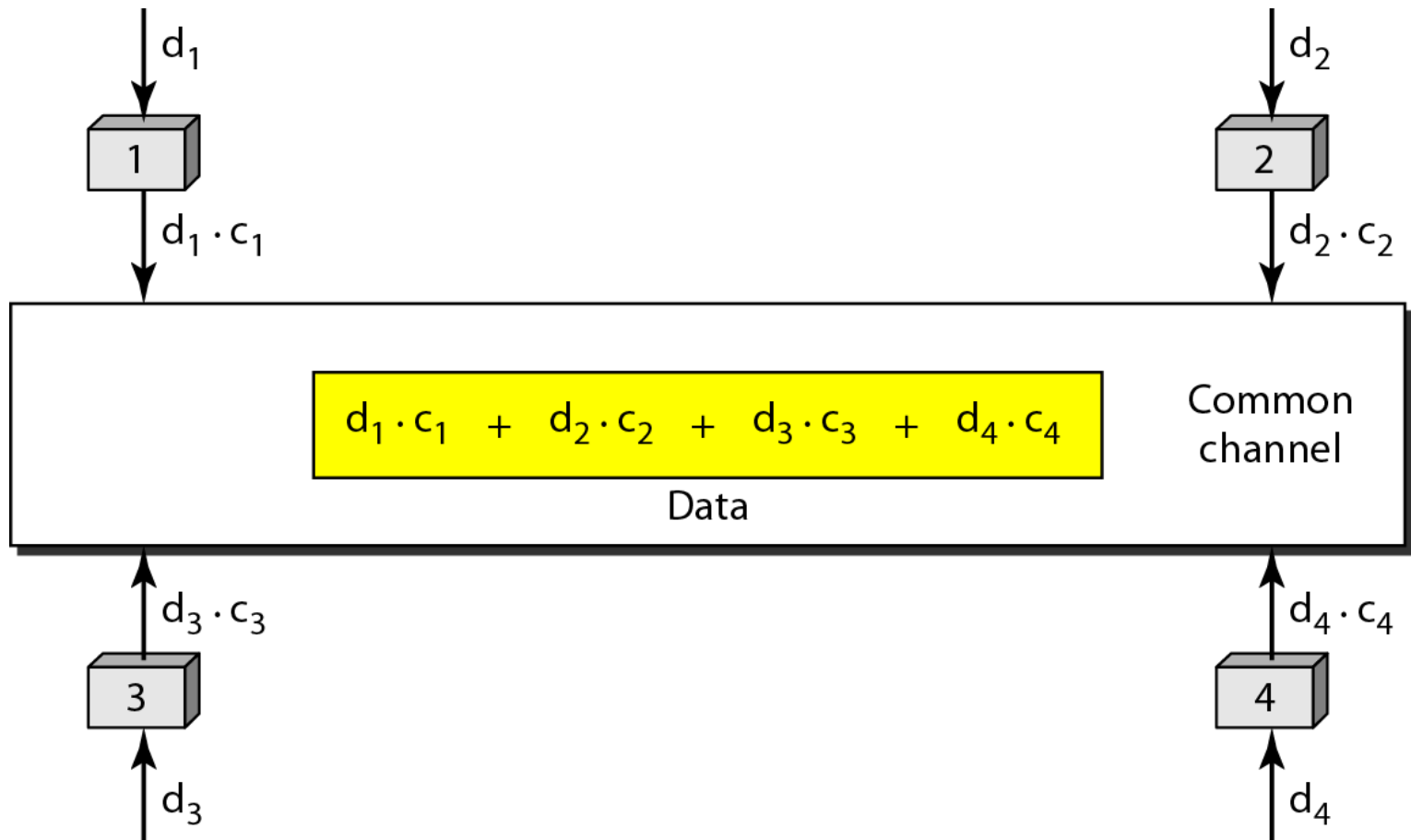
**In TDMA, the bandwidth is just one channel that is timeshared between different stations.**



*Note*

**In CDMA, one channel carries all transmissions simultaneously.**

**Figure 12.23** *Simple idea of communication with code*



## Figure 12.24 *Chip sequences*

$C_1$

[+1 +1 +1 +1]

$C_2$

[+1 -1 +1 -1]

$C_3$

[+1 +1 -1 -1]

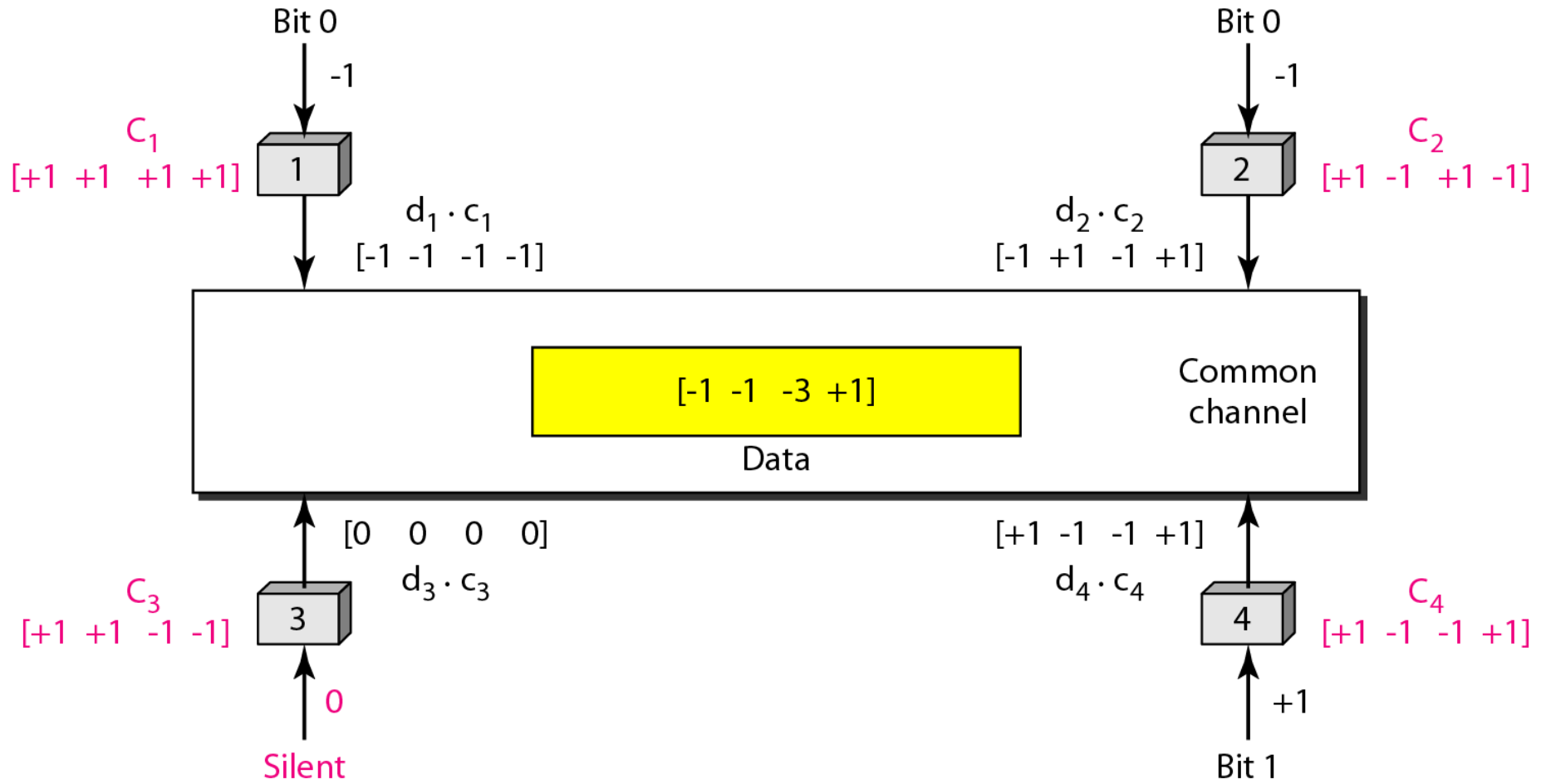
$C_4$

[+1 -1 -1 +1]

## Figure 12.25 *Data representation in CDMA*



**Figure 12.26** *Sharing channel in CDMA*



**Figure 12.27** *Digital signal created by four stations in CDMA*

